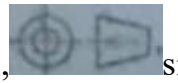
**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY**

**ME 159(TECHNICAL DRAWING) END OF SEMESTEER EXAMS TRIALS.**

TRUE (T) or FALSE (F)

1. A centre line is a thin line. TRUE
2. A soft lead is recommended for drawing. TRUE
3. A Set-square is required for horizontal drawings. FALSE
4. Technically, scale is defined as a ratio of Actual; Drawn. FALSE
5. A regular curve cannot be drawn with a French curve. FALSE
6. Constructional lines are thin lines. TRUE
7. The Path traced by a point attached on a circle that rolls the outside of another circle is called inferior hypotrochoid. FALSE (inferior epitrochoid)
8. The Path traced by a point attached inside a circle that rolls on a straight line is called inferior cycloid. FALSE (inferior trochoid)
9. Epitrochoids, Involute of polygons, Spirals and Helices are space curves. TRUE
10. On the basis of distance between Object and Ray, projections can be termed as perspective and parallel ray projections. TRUE
11. The symbol,  signifies a First Angle Orthographic Projection. TRUE
12. The distance between the centres of two internally tangential arcs is; Cc =R +r. FALSE, rather R - r
13. Hidden outlines are indicated in drawings with thick continuous lines. FALSE
14. Only line type A (thick continuous) and line type D represent edges. TRUE
15. The lengths and spacing of the elements of thick chain lines should differ from those of thin chain lines. FALSE
16. When thin chain lines are used as centre lines, they should cross one another at short portions of the line. FALSE

17. All chain lines should E, should start and finish with a short dash. FALSE

18. Dashed lines should meet at corners and tangent points with dashes. TRUE

19. Where visible line and invisible line coincide in a drawing, the visible line takes precedence. TRUE

20. When invisible line and center line coincide in a drawing, the centre line takes precedence. FALSE

NB; Visible lines takes precedence over hidden lines so when they coincide the hidden lines should be omitted and a hidden line takes precedence over a centerline

21. Outlines of visible edges can be represented by

A. Thin continuous lines

B. Thick short lines

C. Thick chain lines

D. Thick continuous lines

E. None of the above

22. The ratio of thick lines; thin lines recommended for standard drawings is

A. 1;1

B. 1 ;2

C. 1 :3

D. 2 :1

E. 3: 1

23. The following information cannot be found in the title block:

A. Dimension of object
B. Title of drawing
C. Drawn Scale
D. Names and date of drafter. checker, designer
E. Name of firm.

24. A scale 1 :2 of represents:

A. A magnification
B. A reduction
C. A full size scale
D. None of the above

25. The true length of a vertically drawn edge of an object is 50 mm. What is the image length of the object given that the scale of drawing is 1:2

A. 5 mm
B. 10 mm C. 25 mm
D. 50 mm
E. 100 mm

26. Pick odd one out.

A. Parabola
B. Cycloid
C. Ellipse
D. Hyperbola
E. None of the above

27. Circle, Ellipse, Cycloids and Trochoids are all example of:

A. Loci
B. Conic curves
C. Polygons
D. Tangents
E. None of the above

28. The following are properties of orthographic projection except:

A. The rays are orthogonal to the planes of projection
B. Only one side or surface is exposed to the rays
C. The rays are parallel to one another
D. It is a single view projection
E. It is a multi-view projection

29. In a reduction scale drawing, the drawn size is of

A. The same size as the object
B. Reduced size of the object
C. Bigger size of the object
D. Only some of the area is reduced
E. None of the above

30. The two points that determine the accurate position of a line should be_____

A. Reasonably far apart
B. reasonably close to each other
C. either close or apart
D. None

31. The following are all types of projection **except**

A. Parallel projection
B. Perspective projection
C. Oblique projection
D. Axonometric projection
E. None of the above

32. In producing First Angle orthographic projections of an object, the object is assumed to be placed in___first_____ quadrants respectively.

A. Transparent box system
B. Opaque box system
C. Second quadrant
D. Fourth quadrant
E. None of the above

33. The first most important view of an object during Orthographic Projection is captured on

A. Profile plane

B. Horizontal plane

C. Vertical plane

D. Auxiliary plane

E. Horizontal auxiliary plane

34. Figure I shows objects A - E. Select one of the objects from "B" to "E" that would fit exactly into object "A"

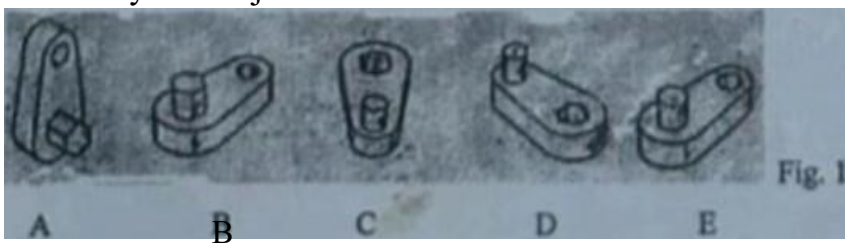


Fig. 1

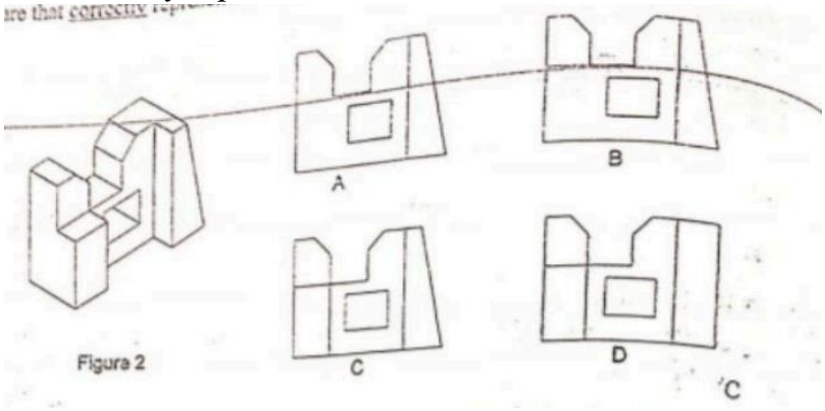
B

ANS; D

35.The following types of projections can be categorized as Axonometric Projections except

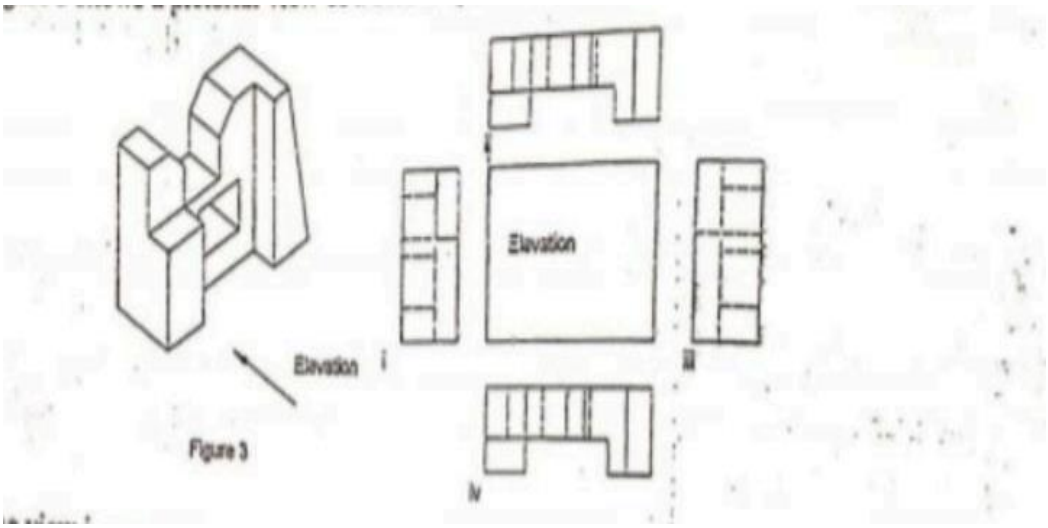
A. Dimetric projection
B. Trimetric projection
C. Isometric projection
D. Perspective projection
E. None

Figure 2 shows a pictorial view of a block. Select from the views on the right side of the figure that correctly represents the front view. c




ANS; C

Figure 3 shows a pictorial view of a block. Questions 37 – 40 are based on this figure




37.The view i, represents;

A. First angle orthographic projection Right Side View

B. First angle orthographic projection Left Side View

C. Third angle orthographic projection Right Side View

D. Third angle orthographic projection Left Side View

38.The view ii, represents;

A. First angle orthographic projection Top View

B. First angle orthographic projection Bottom View

C. Third angle orthographic projection Top View

D. Third angle orthographic projection Bottom View

39.The view iii, represents:

A. First angle orthographic projection Right Side View

B. First angle orthographic projection Left Side View

C. Third angle orthographic projection Right Side View

D. Third angle orthographic projection Left Side View

40.The view iv, represents

A. First angle orthographic projection Top View

B. First angle orthographic projection Bottom View

C. Third angle orthographic projection Top View

D. Third angle orthographic projection Bottom View

SECTION B

Q1. A point P is located from the centre of a circle of radius 40mm as shown in the figure QB1. Construct the path traced by the point if the circle rolls clockwise without slipping through half revolution on a convex surface of radius 100. What is the name of the curve drawn.

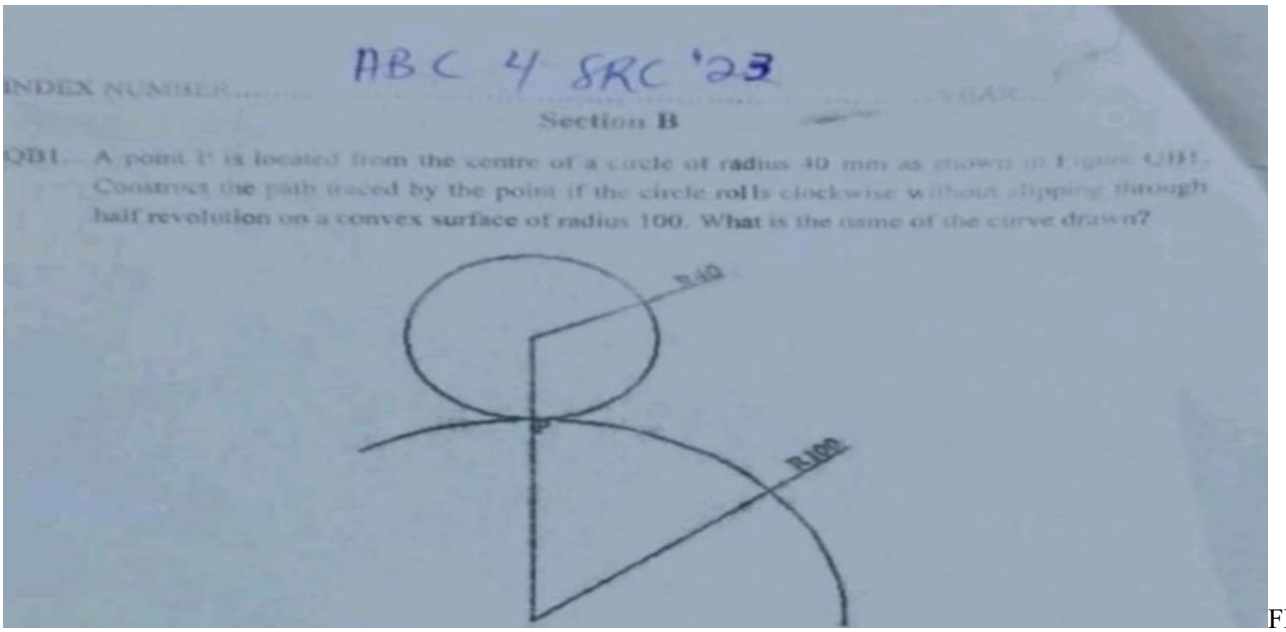
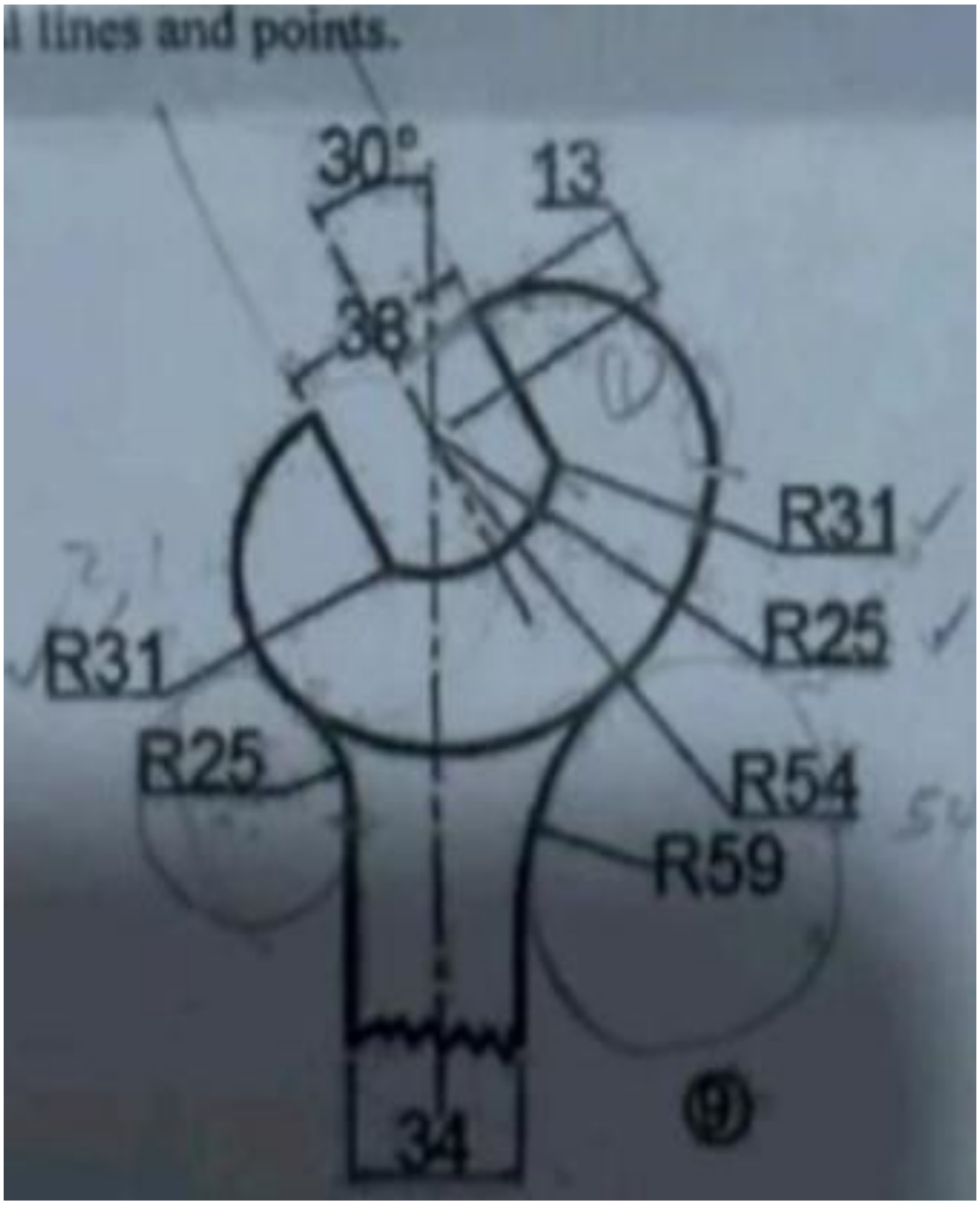


FIG QB1

QB2Draw FULL SIZE the guide Bracket in Figure QB2. Slow clearly all construction lines

l lines and points.



## Section C

C1 Provide the THREE BEST FIRST ANGLE ORTHOGRAPHIC views of the block shown in Figure QC1

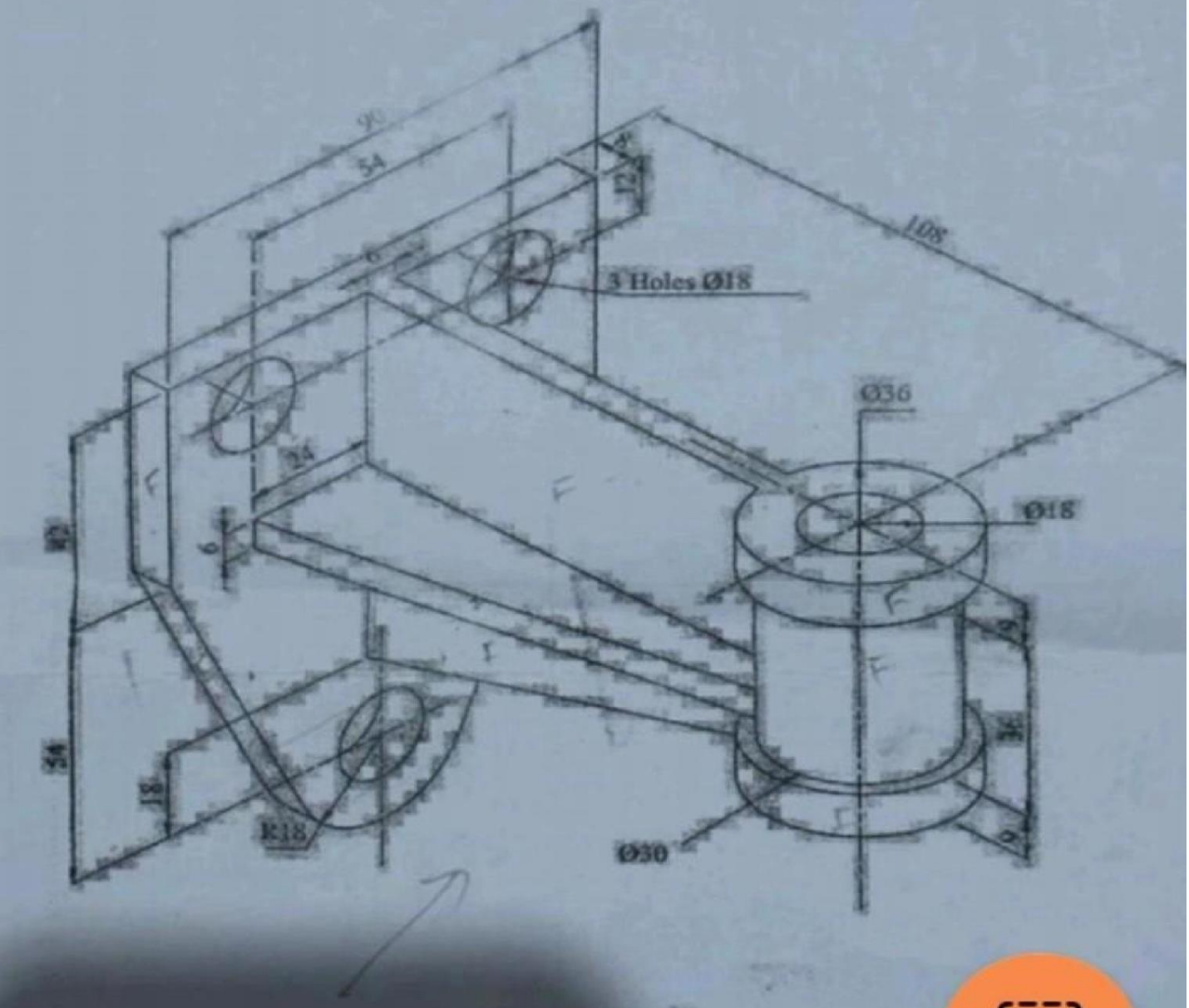
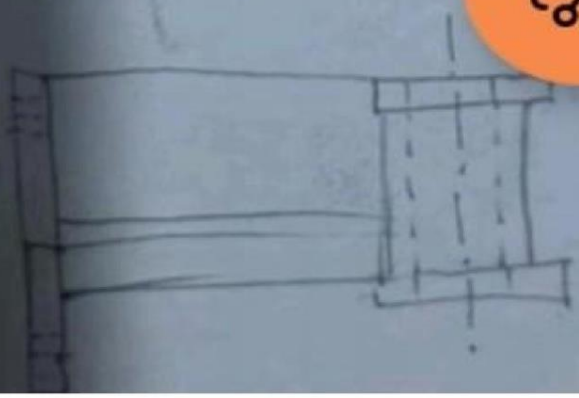



Figure QC1



Aveyire
K. Sunnu
O. Tawiah
S. D. Afrifa